

Hackathon Development Phases:

AgriTech MVP

Timeline: June 30 - July 10, 2026

Strategy: Core functionality first (Marketplace -> Logistics), followed by Integration, and finally Bonus Features (AI/USSD).

Phase 1: Foundation & Scaffold (June 30 - July 1)

Goal: Set up the environments, databases, and project repositories so everyone can work independently without blocking each other.

- **Daniel (Backend):** Set up the Supabase project. Create the initial database schema (Tables: Users [roles: farmer, buyer, driver], Products, Orders). Set up Supabase Authentication.
- **Frederick (Frontend):** Initialize the React + Vite + Tailwind project. Set up GitHub repository and invite the team. Create the basic routing framework (e.g., /farmer-dashboard, /marketplace, /driver-portal).
- **Kojo (AI/Lead):** Initialize the FastAPI Python repository. Connect it to Daniel's Supabase instance. Define the API contract (what JSON the backend will send to Frederick).
- **Afia (PM/Data):** Finalize UI wireframes. Start compiling a spreadsheet of realistic dummy data (Produce names, Tarkwa/Takoradi prices, GPS coordinates for mock farms and buyers).

Phase 2: Core Marketplace - Farmers & Buyers (July 2 - July 4)

Goal: Build the essential "Must-Have" requirement—a farmer uploading produce and a buyer searching/buying it.

- **Daniel:** Build the FastAPI endpoints for POST /products (farmer uploads) and GET /products (buyer searches). Ensure image uploading works.
- **Frederick:** Build the UI for the Farmer Dashboard (form to add produce, upload image, set price). Build the Buyer Marketplace UI (grid of products, basic search bar, and "Add to Cart/Buy" button).
- **Kojo:** Begin developing the XGBoost model in a separate Jupyter Notebook. Start cleaning the dataset Afia provides to train the model on distance vs. price.
- **Afia:** Inject the dummy data directly into Daniel's database so Frederick has real-looking data to display on his UI. Test the farmer upload flow.

Phase 3: Core Logistics - Transport Flow (July 5 - July

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Goal: Connect the driver to the transaction.

- **Frederick:** Build the Transport Portal UI. Integrate Mapbox API to display a map showing the farmer's location and the buyer's destination. Add an "Accept Delivery" button for the driver.
- **Daniel:** Create the Orders and Deliveries tables. Use Supabase's PostGIS to write a query that finds available drivers near the farmer. Build the endpoint to update delivery status (pending -> in-transit -> delivered).
- **Kojo:** Finalize training the XGBoost model. Save the model as a .pkl file. Prepare the Africa's Talking API account for the USSD integration.
- **Afia:** Map out the exact step-by-step script for the live demo video. Ensure the UI flows logically from the user's perspective.

Phase 4: Integration & Bonus Features (July 7 - July 8)

Goal: *Team Sync - "Come together and code"*. Connect the front and back ends, and inject Kojo's AI and Offline bonuses to secure the win.

- **Team Sync (All):** Frederick connects his React frontend to Daniel's FastAPI endpoints. Fix any CORS errors or JSON mismatch issues together.
- **Kojo:** Load the .pkl XGBoost model into the FastAPI server. Create the /api/estimate-cost endpoint. Write the basic USSD script using Africa's Talking API so a farmer can query prices offline.
- **Daniel:** Assist Frederick in connecting WebSockets/Supabase Realtime so the buyer's UI updates instantly when the driver clicks "Picked Up".
- **Frederick:** Consume Kojo's AI endpoint to display the "AI Estimated Delivery Cost" to the buyer before they checkout.
- **Afia:** Conduct brutal end-to-end testing. Try to break the app. Note down all bugs for Phase 5.

Phase 5: Polish & Live Demo Prep (July 9 - July 10)

Goal: Stop writing new features. Fix bugs, make it look professional, and record the winning presentation.

- **Daniel & Kojo:** Bug squashing on the backend. Ensure the server doesn't crash during the demo.
- **Frederick:** UI polish. Fix broken Tailwind layouts, ensure mobile responsiveness, and add loading spinners so the app feels fast.
- **Afia:** Record the live demo video using the finalized platform. Run through the exact workflow required by the judges (Farmer upload -> Buyer search -> Order -> Transport booking).